

KERNEL ESTIMATION OF A DENSITY IN AN UNKNOWN ENDPOINT OF ITS SUPPORT

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Summary: A kernel estimator of the density in an unknown truncation point of the underlying distribution function is investigated. Asymptotic normality is achieved under mild conditions on the distribution.

INTRODUCTION AND MAIN RESULT

Throughout this article let P be a probability measure on the real line with finite left truncation point θ , i.e. there exists $\theta \in \mathbb{R}$ such that $F(\theta) = 0$ where F denotes the distribution function according to P . We assume that for some $\varepsilon > 0$, F is twice differentiable in the interval $I := (\theta, \theta + \varepsilon)$ with F'' uniformly bounded in I and that $f(\theta) := \lim_{t \rightarrow \theta} F'(t) > 0$.

We are concerned with the problem of estimating $f(\theta)$ from a sample $X_1, \dots, X_n$ of independent random variables identically distributed according to P .

The problem of finding a fixed-width confidence interval of level $1 - \alpha$ for the location parameter θ of an exponential

distribution with density

$$f_{\theta, \lambda}(x) := \lambda^{-1} \exp(-(x-\theta)/\lambda) I_{[\theta, \infty)}(x)$$

where λ is also unknown was investigated by Mukhopadhyay (1974). Since there exists no fixed sample procedure, he defined a sequential procedure and proved its asymptotic consistency and efficiency in a certain sense.

For the case of a general distribution Swanepoel and Van Wyk (1981b) defined a histogram type estimator of $f(\theta)$ and proved its strong consistency and asymptotic normality. In a further article Swanepoel and Van Wyk (1981a) investigated a sequential procedure based on their estimator, yielding a fixed-width confidence interval for θ . In the special case of an exponential distribution their procedure has the same asymptotic performance as the one defined by Mukhopadhyay (1974).

A different but nevertheless obvious approach to the problem we are concerned with consists in the application of a kernel density estimator. That means, we estimate $f(\theta)$ by an estimator of the form

$$(1) \quad f_n(Z_{1:n}) := (n\alpha_n)^{-1} \sum_{i=1}^n k((X_i - Z_{i:n})/\alpha_n)$$

where $Z_{i:n}$, $i \in \{1, \dots, n\}$, denotes the i -th ordered value in the sample $X_1, \dots, X_n$, k fulfils the condition $\int_{-\infty}^{\infty} k(x) dx = 1$ and $\alpha_n > 0$, $n \in \mathbb{N}$, tends to zero as n increases.

Estimators of this form have been extensively studied in the literature on nonparametric density estimation (see, for example, the survey articles by Scott *et al.* (1977) and Wertz (1978)). However, consistency of a kernel type estimator of a density is only achieved if the density satisfies certain smoothness conditions. If, in particular, the probability

measure P has finite support $[a, b]$, then it is easy to see that under common conditions on k and α_n :

$$f(t) \quad t \in (a, b)$$

$$(2) \quad f_n(t) \xrightarrow{P} f(t) \int_0^{\infty} k(x) dx, \quad t = a$$

$$f(t) \int_{-\infty}^0 k(x) dx, \quad t = b$$

where $\xrightarrow{P}$ denotes convergence in probability and $f_n(t)$ is defined as in (1) by replacing $Z_{1:n}$ by t .

Thus, at the truncation points of a distribution a kernel estimator of the density usually fails, caused by the fact that the kernel puts weight on the positive as well as on the negative numbers.

This drawback in the otherwise appealing concept of kernel type estimators can be overcome in the following way. Formula (2) suggests the application of one-sided kernels, e.g.

$$\int_0^{\infty} k(x) dx = 1 \quad \text{or} \quad \int_{-\infty}^0 k(x) dx = 1, \quad \text{if the left or the right}$$

truncation point of a distribution is to be dealt with, respectively. If these values are unknown we employ the sample minimum or maximum in f_n and thus are led to estimator (1).

In the present paper we consider in the following the special kernel

$$k_e(x) := \exp(-x) I_{[0, \infty)}(x).$$

Besides its computational advantage this kernel allows us to establish the very useful relation (3) between $f_n(Z_{1:n})$ and $f_n(\theta)$. Moreover, we do not need to sharpen the conditions on F stated at the beginning of this article.

The following representation is obvious

$$(3) \quad f_n(Z_{1:n}) = f_n(\theta) \exp((Z_{1:n} - \theta)/\alpha_n).$$

It is easy to verify that the condition $\alpha_n \xrightarrow[n \in \mathbb{N}]{} \infty$ is necessary as well as sufficient to ensure that $(Z_{1:n} - \theta)/\alpha_n$ tends to zero in probability. This can be achieved by using

$$(4) \quad \Pr\{Z_{1:n} - \theta > \alpha_n \delta\} = [1 - F(\theta + \alpha_n \delta)]^n \\ \leq \exp(-n\alpha_n c \delta) \quad \text{if } \alpha_n \delta \text{ is small}$$

$$\text{where } c := \inf_{x \in (\theta, \theta + \alpha_n \delta)} F'(x).$$

Thus, the condition $\alpha_n \xrightarrow[n \in \mathbb{N}]{} \infty$ is also necessary and sufficient to ensure that $f_n(Z_{1:n})$ and $f_n(\theta)$ are asymptotically equivalent in probability. In particular, if $\alpha_n \xrightarrow[n \in \mathbb{N}]{} \infty$, $f_n(Z_{1:n})$ is a consistent estimator of $f(\theta)$ if and only if $f_n(\theta)$ is!

In order to prove asymptotic normality of $f_n(Z_{1:n})$ we therefore have to compute the mean μ_n and the variance σ_n^2 of $f_n(\theta)$:

$$(5) \quad |\mu_n - f(\theta)| = o(\alpha_n),$$

$$(6) \quad |n\alpha_n \sigma_n^2 - f(\theta)/2| = o(\alpha_n).$$

Integration by parts yields

$$\mu_n = \alpha_n^{-1} \int_{\theta}^{\infty} \exp(-(x-\theta)/\alpha_n) F(dx) \\ = \alpha_n^{-2} \int_{\theta}^{\infty} F(x) \exp(-(x-\theta)/\alpha_n) dx = \alpha_n^{-1} \int_0^{\infty} F(\alpha_n x + \theta) \exp(-x) dx$$

which implies (5), while (6) follows in an analogous way. Moreover, the same arguments show that for any

$j \in \mathbb{N} \int [\alpha_n^{-1} k_e((x-\theta)/\alpha_n)]^j F(dx)$ is finite and of order $O(\alpha_n^{-j+1})$.

Now we are ready to state our main result.

Theorem. Let F be twice differentiable in the interval $I = (\theta, \theta + \varepsilon)$ with F'' uniformly bounded in I and $f(\theta) > 0$. Then, if $n\alpha_n \xrightarrow[n \in \mathbb{N}]{} \infty$

$$(i) \quad n^{\frac{1}{2}} \alpha_n^{\frac{1}{2}} (f_n(Z_{1:n}) - \mu_n) \xrightarrow{D} N(0, f(\theta)/2)$$

and, if $n\alpha_n^3 \xrightarrow[n \in \mathbb{N}]{} 0$

$$(ii) \quad n^{\frac{1}{2}} \alpha_n^{\frac{1}{2}} (f_n(Z_{1:n}) - f(\theta)) \xrightarrow{D} N(0, f(\theta)/2)$$

where $N(a, b)$ denotes the normal distribution with mean a and variance b and $\xrightarrow{D}$ weak convergence.

Proof: For any $\delta > 0$

$$\begin{aligned} & \Pr\{n^{\frac{1}{2}} \alpha_n^{\frac{1}{2}} (\exp((Z_{1:n} - \theta)/\alpha_n) - 1) \geq \delta\} \\ & \leq \Pr\{(Z_{1:n} - \theta)/\alpha_n \geq \tilde{\delta} n^{-\frac{1}{2}} \alpha_n^{-\frac{1}{2}}\} \leq \exp(-n^{\frac{1}{2}} \alpha_n^{\frac{1}{2}} \tilde{\delta} c) \end{aligned}$$

by (4) if n is large, where $\tilde{\delta}$ is appropriately chosen. Now (i) follows from (3), (5), (6), Slutsky's Theorem and the Berry-Esséen Theorem for independent identically distributed random variables. (ii) is immediate from (i), (5) and Slutsky's Theorem.

REMARKS

The above theorem states that the speed of convergence of $f_n(Z_{1:n})$ to $f(\theta)$ is of order $(n\alpha_n)^{-\frac{1}{2}}$ where $n\alpha_n \rightarrow 0$ $n \in \mathbb{N}$

which is the same rate as for the estimator defined by Swanepoel and Van Wyk (1981b). On the other hand it is clear that under more restrictive conditions on F and k one obtains by (1) estimators which converge to $f(\theta)$ at a faster rate.

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